SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50

Annexure A

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

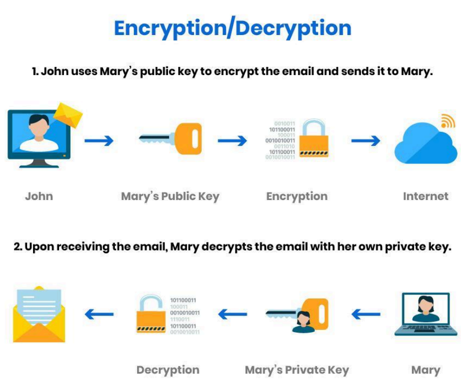
Signature of the student:

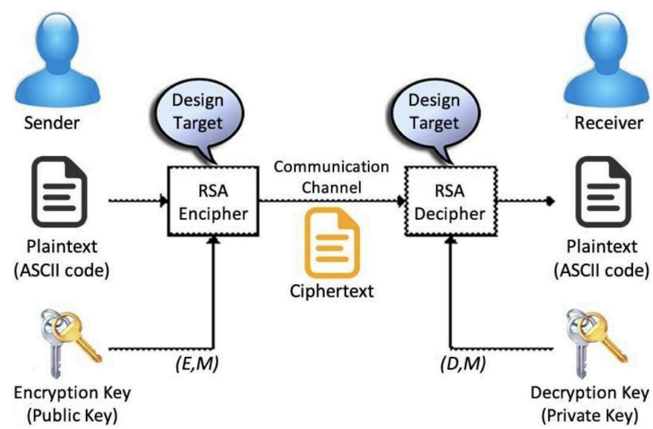
Annexure A

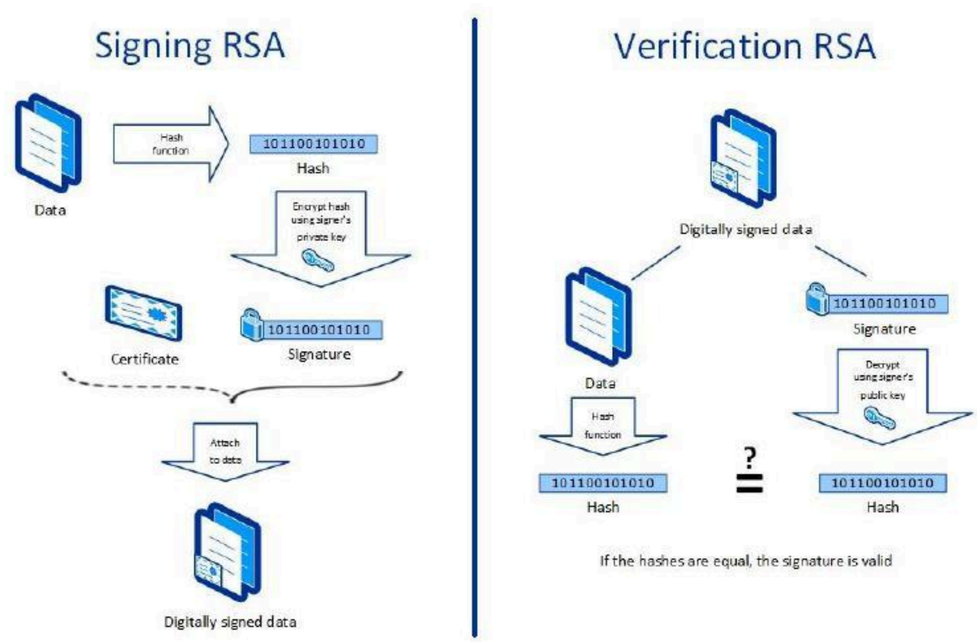
Scenario-based

4. Public Key Cryptography for Email Security

RSA Email Security Architecture







Scenario

An organization wants to protect confidential emails exchanged between employees using the **RSA public key cryptography algorithm**. RSA ensures that only the intended recipient can read the email while also supporting authentication through digital signatures.

A. How RSA is Used for Encryption and Decryption

Step 1: Key Generation

Each employee generates two keys:

- **Public Key** – Shared with everyone.
- **Private Key** – Kept secret by the owner.

Example:

Alice

Public Key → Shared

Private Key → Secret

Step 2: Encryption

When Bob wants to send an email to Alice:

1. Bob writes the email.
2. Bob encrypts the email using **Alice's Public Key**.
3. The encrypted email is sent over the Internet.

Since only Alice has the corresponding private key, no one else can read the email.

Step 3: Decryption

1. Alice receives the encrypted email.
2. Alice decrypts it using her **Private Key**.
3. The original email is recovered.

RSA Email Communication Process

Alice

Generates Public & Private Keys

Public Key

Bob Receives

Alice's Public Key

Write Email Message

Encrypt using Alice's Public Key

Encrypted Email (Ciphertext)

Internet / Network

Alice Receives

|
▼

Decrypt using Alice's Private Key

|
▼

Original Email Displayed

Digital Signature

RSA can also verify the sender's identity.

Process:

- Sender signs the email using their **Private Key**.
- Receiver verifies the signature using the sender's **Public Key**.

Benefits:

- Authentication
- Integrity
- Non-repudiation

B. Role of Key Distribution and Management

Key Distribution

The public key must be distributed securely.

Methods include:

- Certificate Authority (CA)
- Public Key Infrastructure (PKI)
- Digital Certificates

- Secure Key Servers

Without proper distribution, attackers may replace the public key with a fake one (Man-in-the-Middle Attack).

Key Management

Good key management includes:

- Secure storage of private keys
- Periodic key renewal
- Key revocation if compromised
- Secure backup and recovery
- Certificate validation

Importance

Proper key management:

- Prevents unauthorized access
- Protects confidential information
- Ensures secure communication
- Prevents impersonation attacks
- Maintains authentication and integrity

Advantages of RSA

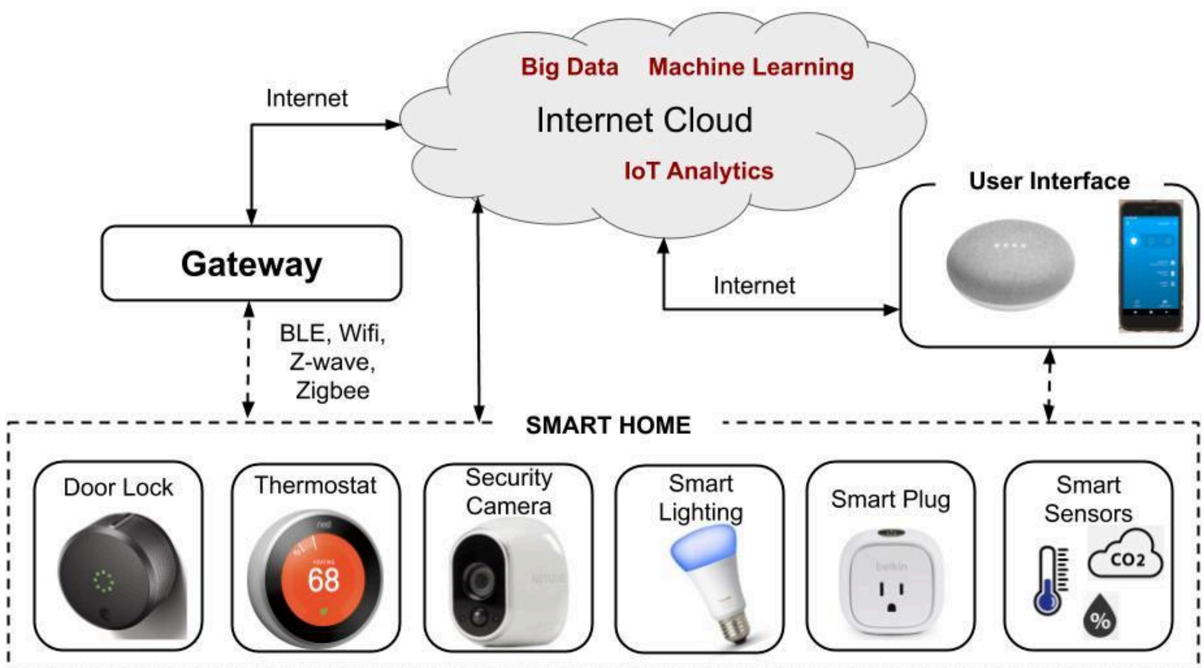
- Secure key exchange
- Strong authentication
- Digital signatures
- Confidential communication
- Widely used in SSL/TLS, Email Security (PGP, S/MIME), Banking, VPNs

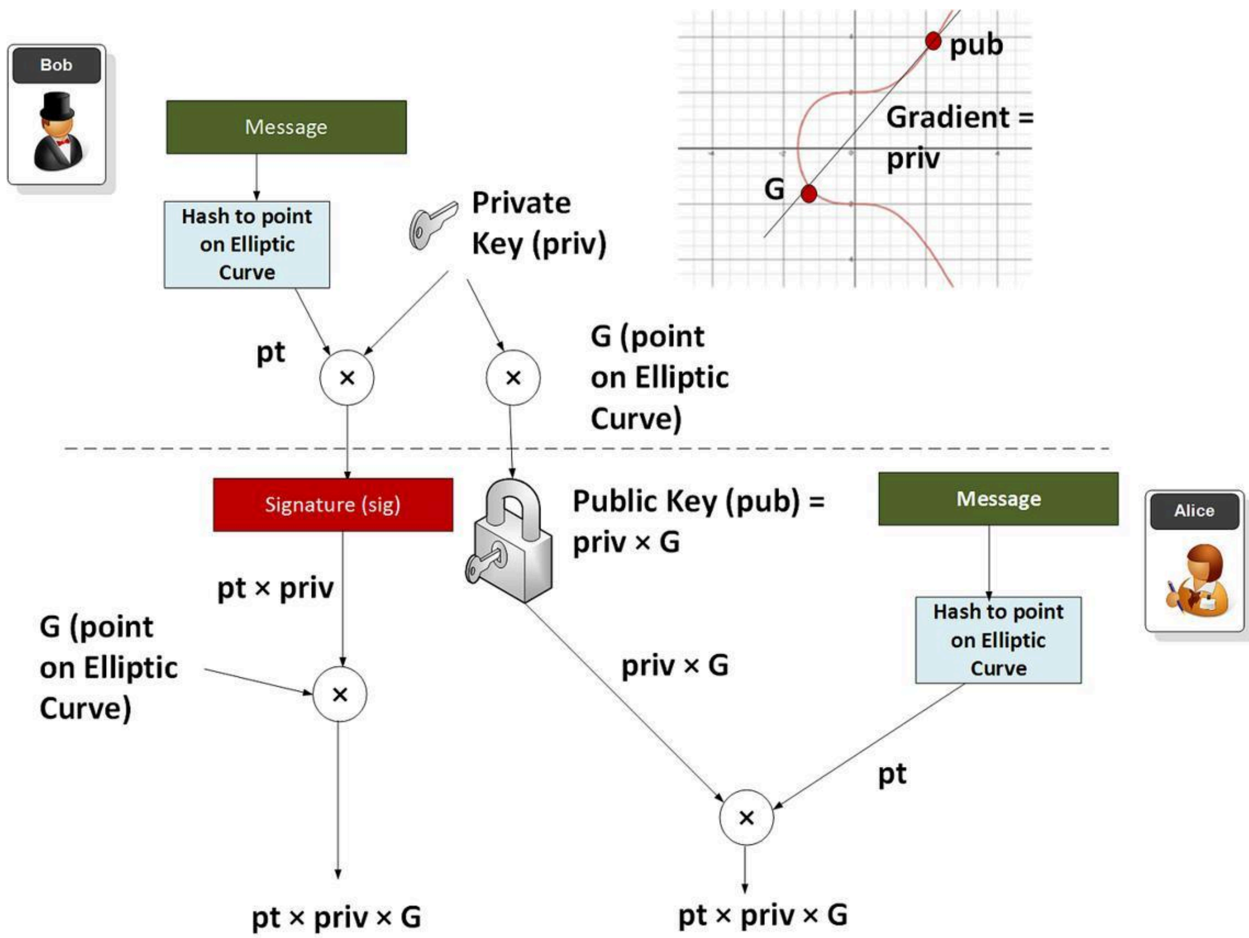
Conclusion

RSA enables secure email communication by encrypting messages with the receiver's public key and decrypting them with the receiver's private key. Proper key distribution and management ensure confidentiality, authentication, integrity, and protection against cyberattacks.

5. Cryptography for IoT Devices

IoT Security with ECC





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Scenario

A smart home consists of multiple IoT devices such as:

- Smart Lights
- Smart Door Locks
- Smart Cameras
- Motion Sensors
- Smart Thermostats
- Smart Watches

These devices continuously exchange sensitive information over wireless networks.

The designer must choose between **RSA** and **Elliptic Curve Cryptography (ECC)**.

A. Constraints of IoT Devices

IoT devices have limited hardware resources.

Major Constraints

1. Limited Processing Power

Most IoT devices use low-speed microcontrollers.

Heavy cryptographic algorithms increase execution time.

2. Limited Memory

Typical RAM ranges from

- 32 KB
- 64 KB
- 128 KB

Large cryptographic keys consume significant memory.

3. Low Battery Power

Most sensors operate on batteries.

Complex encryption algorithms reduce battery life.

4. Limited Storage

Flash memory is limited.

Smaller cryptographic keys reduce storage requirements.

5. Low Network Bandwidth

Wireless communication must be efficient.

Large keys increase communication overhead.

6. Real-Time Communication

IoT devices must respond quickly.

Encryption and decryption should introduce minimal delay.

B. Recommended Cryptographic Approach

Elliptic Curve Cryptography (ECC)

ECC provides security equivalent to RSA while using much smaller key sizes.

RSA vs ECC Comparison

Feature	RSA	ECC
Key Size	2048 bits	256 bits
Security	High	Very High
Encryption Speed	Moderate	Faster
Memory Usage	High	Low

Power Consumption	High	Low
Storage Requirement	High	Low
Battery Efficiency	Poor	Excellent
Suitable for IoT	No	Yes

Why ECC is Better

Smaller Keys

ECC uses much smaller keys while maintaining the same security level.

Example:

RSA 2048-bit Security ≈ ECC 256-bit Security

Faster Computation

Smaller keys reduce mathematical computations.

Result:

- Faster encryption
- Faster decryption

Low Power Consumption

ECC requires fewer CPU cycles.

Battery-powered devices last longer.

Better Performance

ECC offers:

- Low latency
- Low memory usage
- High throughput

Real-World Applications

ECC is widely used in:

- Smart Home Devices
- IoT Sensors
- Smart Watches
- Mobile Banking
- Bluetooth Security
- SSL/TLS
- Cryptocurrency (Bitcoin)

Overall Recommendation

For smart home IoT systems, **Elliptic Curve Cryptography (ECC)** is the preferred choice because it provides strong security with smaller keys, lower power consumption, faster computation, and efficient use of memory and bandwidth. These characteristics make ECC far more suitable than RSA for resource-constrained IoT environments.